

Assignment Code: DS-AG-005

Statistics Basics| Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 200

Question 1: What is the difference between descriptive statistics and inferential statistics? Explain with examples.

Answer:

1. Descriptive Statistics

◇ Definition:

Descriptive statistics are used to **summarize, organize, and describe** the main features of a dataset.

They deal only with the data you have — **no predictions or generalizations** beyond it.

◇ Purpose:

To **present information** about a dataset in a simple, understandable form.

◇ Common Tools:

- Measures of **central tendency** → Mean, Median, Mode
- Measures of **dispersion** → Range, Variance, Standard Deviation
- **Graphs and charts** → Bar chart, Pie chart, Histogram.

Example:

OptiConnect collected satisfaction ratings from 200 customer calls.

Metric	Value
Mean satisfaction score	4.3
Median satisfaction score	4.5
Standard deviation	0.6

2. Inferential Statistics

◇ Definition:

Inferential statistics use sample data to **make conclusions or predictions** about a **larger population**.

It helps you go beyond the observed data.

◇ **Purpose:**

To **infer** (estimate, predict, or test hypotheses) about the population based on sample data.

◇ **Common Tools:**

- **Confidence intervals**
- **Hypothesis testing** (t-test, chi-square test, ANOVA)
- **Regression analysis**
- **Correlation analysis**

◇ **Example:**

OptiConnect wants to know if a **new training program** for agents improved satisfaction ratings.

You collect:

- Before training: mean = 4.1 (sample of 100 calls)
- After training: mean = 4.5 (sample of 100 calls)

You perform a **t-test** to check if the difference is statistically significant.

Aspect	Descriptive Statistics	Inferential Statistics
Purpose	Summarize and describe data	Make inferences or predictions
Scope	Only the dataset in hand	The larger population
Techniques	Mean, median, mode, SD, charts	Hypothesis tests, confidence intervals, regression
Example	Average satisfaction score = 4.3	“Customer satisfaction has increased significantly after training”

Question 2: What is sampling in statistics? Explain the differences between random and stratified sampling.

Answer:

Definition:

Sampling is the process of selecting a **subset (sample)** from a **larger group (population)** in order to draw conclusions about the entire population.

◇ **Why Sampling is Needed:**

- Studying an **entire population** (called a *census*) is often **impractical, time-consuming, and expensive**.
- So, we study a **smaller group (sample)** that **represents the population**.

◇ **Example:**

OptiConnect Solutions handles **50,000 customer calls** per month.

You can't analyze every call manually, so you **select 1,000 calls** as a **sample** to study patterns like satisfaction, resolution rate, etc.

A. Random Sampling

◇ **Definition:**

Every individual (or record) in the population has an **equal chance** of being selected.

There is **no bias** in how samples are chosen.

◇ **Example:**

You have 50,000 call records.

You use Power BI, Excel, or Python to randomly select 1,000 calls — without considering the department, date, or agent.

Each call has the same probability (1 in 50,000) of being picked.

◇ **Advantages:**

- Simple and unbiased
- Easy to understand and implement

◇ **Disadvantages:**

- May not represent **subgroups** (e.g., different departments) equally
- Random chance might over- or under-sample certain groups

Stratified Sampling

◇ **Definition:**

The population is divided into distinct subgroups (strata) that share similar characteristics, and random samples are drawn from each stratum in proportion to their size.

◇ **Example:**

At OptiConnect, calls come from 3 departments:

Department	Total Calls	% of Total
Technical Support	25,000	50%
Billing	15,000	30%
Sales	10,000	20%

You want a **sample of 1,000 calls**, but you want it to represent all departments fairly.

So, you select:

- 500 from Technical Support
- 300 from Billing
- 200 from Sales

Each group is **sampled randomly within its own stratum**.

◇ **Advantages:**

- Ensures **representation** of all key subgroups
- Reduces **sampling error** compared to simple random sampling

◇ **Disadvantages:**

- Requires prior knowledge of population structure
- Slightly more complex to implement

Question 3: Define mean, median, and mode. Explain why these measures of central tendency are important.

Answer:

Mean, median, and mode are measures of central tendency, which are single values that represent the center or typical value of a dataset. Each provides a different perspective on the data, and understanding their differences is essential for accurate analysis.

Definitions

Mean

Also known as the average, the mean is calculated by summing all the values in a dataset and dividing by the number of values.

- **Formula:**

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Mean = Sum of all values / Number of values

- **Example:** For the dataset {2, 4, 6, 8, 10}, the mean is

$$\frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$$

2+4+6+8+10=30
30/5=6

Median

The median is the middle value of a dataset that has been ordered from lowest to highest.

- If the dataset has an **odd number of values**, the median is the single middle number. For the dataset {2, 4, 8, 10, 19}, the median is 8.
- If the dataset has an **even number of values**, the median is the average of the two middle numbers. For the dataset {2, 4, 8, 10, 19, 21}, the median is

$$\frac{8 + 10}{2} = 9$$

8+10=18
18/2=9

Mode

The mode is the value that appears most frequently in a dataset.

- A dataset can have one mode (unimodal), more than one mode (bimodal or multimodal), or no mode at all if every value appears only once.
- **Example:** In the dataset {3, 3, 6, 9, 16, 16, 16, 27, 27, 37, 48}, the mode is 16 because it appears more often than any other number.

Importance of these measures

Different types of datasets and analytical goals require different measures of central tendency. The importance of each measure is determined by how it handles different data characteristics.

- **Handling outliers:** The mean is sensitive to outliers, or extreme values, which can skew the average and make it unrepresentative of the typical value. The median, by contrast, is not affected by outliers, making it a better measure for skewed data like household income or housing prices.
- **Data type:** The mode is the only measure that can be used with categorical data, such as determining the most popular brand of a product or the most common color. The mean and median require numerical data.
- **Identifying frequency:** The mode is valuable when you need to know the most common item or value. For instance, a retailer might use the mode to determine which item sells most often to inform their inventory.
- **Distribution shape:** For a perfectly symmetrical distribution, the mean, median, and mode will all be the same, giving a clear picture of the data's center. For skewed distributions, comparing the three measures reveals the direction and extent of the skew.
- **Overall understanding:** Using all three measures together provides a more complete picture of the data's characteristics. For example, knowing the mean, median, and mode salaries at a company can reveal whether most employees are clustered around the average or if a few high salaries are skewing the mean.

Question 4: Explain skewness and kurtosis. What does a positive skew imply about the data?

Answer:

Skewness

Skewness measures the asymmetry of a data distribution.

- **Zero skewness:** A distribution with zero skewness is perfectly symmetrical, like a normal or "bell-shaped" curve. In this case, the mean, median, and mode are all equal.
- **Negative skewness (left-skewed):** The tail on the left side of the distribution is longer or fatter than the right side. This means that the majority of the data is concentrated on the right, but a few extremely low values pull the mean to the left. In this case, $\text{Mean} < \text{Median} < \text{Mode}$.
- **Positive skewness (right-skewed):** The tail on the right side of the distribution is longer or fatter. This happens when the majority of data points are concentrated on the left, but a few unusually high values pull the mean to the right.

Kurtosis

Kurtosis measures the "tailedness" of a distribution, indicating how heavy or light the tails are compared to a normal distribution. It provides insight into the probability of outliers or extreme values. The "excess kurtosis" compares a distribution's kurtosis to that of a normal distribution, which has a kurtosis of 3 (or an excess kurtosis of 0).

- **Mesokurtic:** Distributions with an excess kurtosis of 0 are mesokurtic. This is characteristic of a normal distribution.
- **Leptokurtic (positive excess kurtosis):** These distributions have fatter, heavier tails and a sharper peak than a normal distribution. A high kurtosis value suggests a high probability of extreme values (outliers).
- **Platykurtic (negative excess kurtosis):** These distributions have thinner, lighter tails and a lower peak. They produce fewer and less extreme outliers than a normal distribution.

What a positive skew implies

A positively skewed distribution (or right-skewed distribution) has a longer tail on the right side of the graph. This implies the following about the data:

- **Presence of high-value outliers:** There are a few very large data points that disproportionately affect the mean, pulling it up towards the right.
- **Mean > Median > Mode:** The relationship between the measures of central tendency is $\text{Mean} > \text{Median} > \text{Mode}$. The mode is the most frequent value, which is on the left where the data is concentrated. The median is the middle value. The mean is the highest value because it is pulled towards the high-value outliers in the right tail.
- **Concentration of data:** Most of the data values are clustered on the left side of the distribution around the mode, with fewer observations scattered in the longer, right-side tail.
- **Real-world example: Income distribution.** In many countries, the income distribution is positively skewed. The majority of the population earns lower to middle-range incomes (clustered on the left), while a small number of people in the high-income bracket earn extremely high salaries, pulling the average income (the mean) up.

Question 5: Implement a Python program to compute the mean, median, and mode of a given list of numbers.

```
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

(Include your Python code and output in the code box below.)

Answer:

Paste your code and output inside the box below:

```
import collections
import statistics

numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]

def calculate_mean(data):
    return sum(data) / len(data)

def calculate_median(data):
    sorted_data = sorted(data)
    n = len(sorted_data)
    if n % 2 == 1:
        # Odd number of elements
        return sorted_data[n // 2]
    else:
        # Even number of elements
        mid1 = sorted_data[n // 2 - 1]
        mid2 = sorted_data[n // 2]
        return (mid1 + mid2) / 2

def calculate_mode(data):
    # Using collections.Counter to find the most common element(s)
```

```
counts = collections.Counter(data)
max_count = max(counts.values())
mode_values = [key for key, value in counts.items() if value == max_count]
return mode_values

mean = calculate_mean(numbers)
median = calculate_median(numbers)
mode = calculate_mode(numbers)

print(f"The list of numbers is: {numbers}")
print("-" * 30)
print(f"Mean: {mean}")
print(f"Median: {median}")
print(f"Mode: {mode}")

The list of numbers is: [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
-----
Mean: 19.6
Median: 19
Mode: [12, 19, 24]
```

Question 6: Compute the covariance and correlation coefficient between the following two datasets provided as lists in Python:

```
list_x = [10, 20, 30, 40, 50] list_y
= [15, 25, 35, 45, 60]
```

(Include your Python code and output in the code box below.)

Answer:

Paste your code and output inside the box below:


```
import math

list_x = [10, 20, 30, 40, 50]
list_y = [15, 25, 35, 45, 60]

def calculate_mean(data):
    """Calculates the mean (average) of a list of numbers."""
    return sum(data) / len(data)

def calculate_covariance(x, y):
    """Calculates the sample covariance between two lists of numbers."""
    if len(x) != len(y) or len(x) < 2:
        return None

    n = len(x)
    mean_x = calculate_mean(x)
    mean_y = calculate_mean(y)

    covariance = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(n)) / (n - 1)
    return covariance

def calculate_std_dev(data):
    """Calculates the sample standard deviation of a list of numbers."""
    if len(data) < 2:
        return None

    n = len(data)
    mean = calculate_mean(data)
    variance = sum((x - mean) ** 2 for x in data) / (n - 1)
    std_dev = math.sqrt(variance)
    return std_dev

def calculate_correlation_coefficient(x, y):
    """Calculates the Pearson correlation coefficient between two lists of numbers."""
    cov = calculate_covariance(x, y)
    std_dev_x = calculate_std_dev(x)
    std_dev_y = calculate_std_dev(y)

    if std_dev_x == 0 or std_dev_y == 0:
        return None

    correlation = cov / (std_dev_x * std_dev_y)
```

```
return correlation

covariance = calculate_covariance(list_x, list_y)
correlation_coefficient = calculate_correlation_coefficient(list_x, list_y)

print(f"Dataset X: {list_x}")
print(f"Dataset Y: {list_y}")
print("-" * 30)
print(f"Covariance: {covariance}")
print(f"Correlation Coefficient: {correlation_coefficient}")

Dataset X: [10, 20, 30, 40, 50]
Dataset Y: [15, 25, 35, 45, 60]
-----
Covariance: 281.25
Correlation Coefficient: 0.9942702718134764
```

Question 7: Write a Python script to draw a boxplot for the following numeric list and identify its outliers. Explain the result:

```
data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]
```

(Include your Python code and output in the code box below.)

Answer:

```
import matplotlib.pyplot as plt
import numpy as np

data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

# Calculate quartiles and IQR to identify outliers
q1 = np.percentile(data, 25)
q3 = np.percentile(data, 75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

outliers = [x for x in data if x < lower_bound or x > upper_bound]
```

```
print(f"The list of numbers is: {data}")
print("-" * 30)
print(f"First Quartile (Q1): {q1}")
print(f"Third Quartile (Q3): {q3}")
print(f"Interquartile Range (IQR): {iqr}")
print(f"Lower Bound: {lower_bound}")
print(f"Upper Bound: {upper_bound}")
print(f"Identified Outliers: {outliers}")
print("-" * 30)

# Create the boxplot
fig, ax = plt.subplots()
ax.boxplot(data)
ax.set_title("Boxplot for the given data")
ax.set_ylabel("Value")
ax.set_xticks([1])
ax.set_xticklabels(['Data'])

plt.show()
```

The list of numbers is: [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

First Quartile (Q1): 18.75
Third Quartile (Q3): 23.75
Interquartile Range (IQR): 5.0
Lower Bound: 11.25
Upper Bound: 31.25
Identified Outliers: [35]

Question 8: You are working as a data analyst in an e-commerce company. The marketing team wants to know if there is a relationship between advertising spend and daily sales.

- Explain how you would use covariance and correlation to explore this relationship.
- Write Python code to compute the correlation between the two lists:

```
advertising_spend = [200, 250, 300, 400, 500]  daily_sales
= [2200, 2450, 2750, 3200, 4000]
```

(Include your Python code and output in the code box below.)

Answer:

As a data analyst, I would use **covariance** and **correlation** to explore the relationship between advertising spend and daily sales. These statistical measures help quantify how two variables move together.

- **Covariance**

Covariance measures the directional relationship between two variables. A positive covariance indicates that as advertising spend increases, daily sales also tend to increase. A negative covariance would suggest the opposite—that as one increases, the other decreases. The main limitation of covariance is that its magnitude is not standardized, making it difficult to interpret the strength of the relationship.

- **Correlation**

Correlation, specifically the Pearson correlation coefficient, is a more useful measure. It not only indicates the direction of the relationship (like covariance) but also quantifies its **strength** on a standardized scale from -1 to +1.

- A coefficient of **+1** would mean a perfect positive linear relationship: every increase in advertising spend corresponds to a proportional increase in sales.
- A coefficient of **-1** would mean a perfect negative linear relationship.
- A coefficient of **0** would mean no linear relationship exists between the two variables.

By calculating the correlation coefficient for the given data, I can provide the marketing team with a single, easily interpretable number that tells them how strongly advertising spend and daily sales are related. A high positive value would support their strategy of increasing ad spend to boost sales.

```
import numpy as np

# Provided datasets
advertising_spend = [200, 250, 300, 400, 500]
daily_sales = [2200, 2450, 2750, 3200, 4000]

# Calculate the correlation matrix
# The correlation coefficient between the two lists is the off-diagonal element
correlation_matrix = np.corrcoef(advertising_spend, daily_sales)

# Extract the correlation coefficient
correlation_coefficient = correlation_matrix[0, 1]

print(f"Advertising Spend: {advertising_spend}")
print(f"Daily Sales: {daily_sales}")
print("-" * 30)
```

```
print(f"Correlation Coefficient: {correlation_coefficient}")
```

Advertising Spend:

Daily Sales:

Correlation Coefficient: 0.999517173950478

Question 9: Your team has collected customer satisfaction survey data on a scale of 1-10 and wants to understand its distribution before launching a new product.

- Explain which summary statistics and visualizations (e.g. mean, standard deviation, histogram) you'd use.
- Write Python code to create a histogram using Matplotlib for the survey data:

`survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]` (Include your Python code and output in the code box below.)

Answer:

To understand the distribution of customer satisfaction survey data, a data analyst should use a combination of summary statistics and visualizations.

Summary Statistics

- **Mean:** The average score gives a central value for the overall satisfaction. A high mean suggests generally positive feedback.
- **Median:** As the middle score, the median is less affected by extreme values than the mean. Comparing the mean and median can reveal if the data is skewed by a few very high or low scores.
- **Mode:** The mode, or most frequent score, shows the most common level of satisfaction.
- **Standard Deviation:** This measures the amount of variation or spread in the scores. A small standard deviation indicates that scores are clustered closely around the mean, suggesting consistent feedback. A large standard deviation indicates a wider spread of scores, implying greater inconsistency in feedback.

Visualizations

- **Histogram:** A histogram is a bar chart that displays the frequency of each score. For this data, with a scale of 1 to 10, a histogram will visually reveal the shape of the distribution.

- **Skewness:** It helps identify if the data is skewed. For example, a left-skewed distribution (longer tail to the left) would mean most customers are highly satisfied, but a few are very dissatisfied.
- **Spread:** It shows the spread of the data, reinforcing the information from the standard deviation.
- **Central Tendency:** It visually shows the most common scores and where the center of the data lies.
- **Multimodality:** It can reveal if the data has more than one peak, which could indicate distinct segments of customers with very different satisfaction levels.

```
import matplotlib.pyplot as plt
import numpy as np
```

```
# Survey data
```

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]
```

```
# Create the histogram
```

```
plt.figure(figsize=(10, 6))
```

```
plt.hist(survey_scores, bins=np.arange(3.5, 11.5, 1), edgecolor='black', alpha=0.7)
```

```
# Add title and labels
```

```
plt.title('Distribution of Customer Satisfaction Survey Scores (1-10)', fontsize=16)
```

```
plt.xlabel('Satisfaction Score', fontsize=12)
```

```
plt.ylabel('Frequency', fontsize=12)
```

```
plt.xticks(np.arange(4, 11))
```

```
plt.grid(axis='y', alpha=0.5)
```

```
# Show the plot
```

```
plt.show()
```

```
# Print summary statistics
```

```
mean_score = np.mean(survey_scores)
```

```
median_score = np.median(survey_scores)
```

```
mode_score = max(set(survey_scores), key=survey_scores.count)
```

```
std_dev = np.std(survey_scores)
```

```
print(f"Mean Score: {mean_score:.2f}")
```

```
print(f"Median Score: {median_score}")
```

```
print(f"Mode Score: {mode_score}")
```

```
print(f"Standard Deviation: {std_dev:.2f}")
```

Mean Score: 7.47

Median Score: 7.0

Mode Score: 7

Standard Deviation: 1.69